INCH-POUND
MIL-PRF-39002/1G
23 August 2010
SUPERSEDING
MIL-PRF-39002/1F
14 May 2004

PERFORMANCE SPECIFICATION

RESISTOR, VARIABLE, WIREWOUND, SEMI-PRECISION, STYLE RK09

This specification is approved for use by all Departments
And Agencies of the Departments of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39002.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for a variable, wirewound, semi-precision resistors. These resistors are capable of full load operation at a maximum ambient temperature of 85°C, and suitable for continuous operation. These resistors have a tolerance of ±5 percent.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:

M39002/01	-	0001
Performance specification number		Coded dash number

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information are as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-39002 - Resistor, Variable, Wire-Wound, Semi-Precision, General Specification for.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to Resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 5905

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202 - Test Method Standard, Electronic and Electrical Component Parts.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39002.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Power rating. The power rating shall be 1.5 watts based on full load operation at an ambient temperature of 85°C.

3.4 Resistance value. Resistance value shall be in accordance with table I.

3.5 Shaft length. Shaft length shall be in accordance with table I.

3.6 Shaft style. Shaft style shall be in accordance with table I.

3.7 Bushing. The bushing shall be in accordance with table I.

3.8 Nominal resolution. Nominal resolution shall be in accordance with table I.

3.9 Rated working voltage. Rated working voltage shall be in accordance with table I.

3.10 Mechanical travel. The mechanical travel shall be $300^{O} \pm 5^{O}$.

3.11 Actual effective electrical travel. The actual effective electrical travel shall be $300^{O} \pm 5^{O}$.

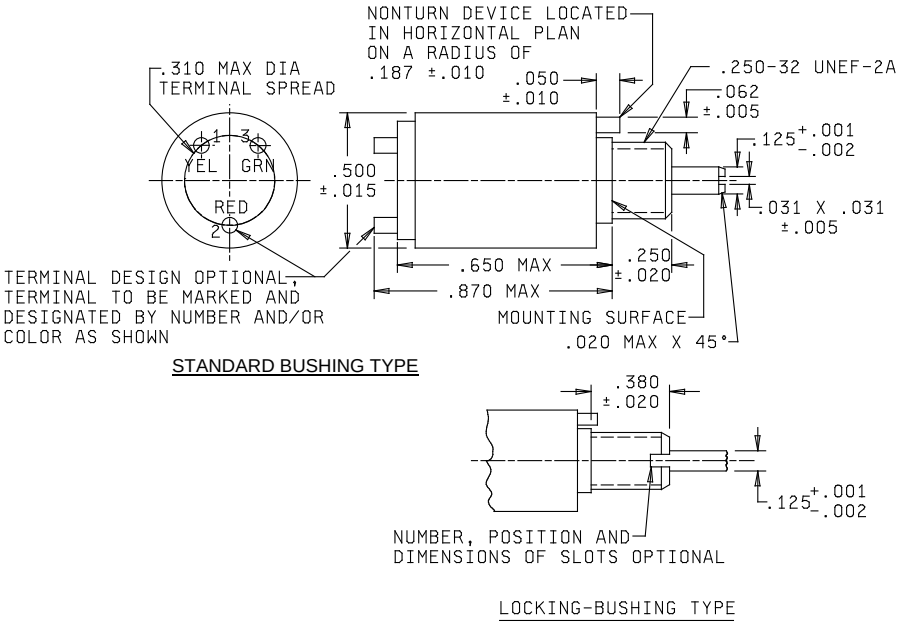
3.12 End resistance. The end resistance shall not exceed 3 percent of the nominal total resistance value or 1 ohm, whichever is greater.

3.13 Electrical output. The electrical output (in terms of percent of applied voltage) is linear with respect to the angular position of the contact arm.

3.14 Hardware. The mounting nut supplied shall be 0.0625 inch (1.588 mm) thick and shall measure 0.3125 inch (7.938 mm) across the hexagonal flats. For locking bushings, the locking nut shall be 0.125 inch (3.18 mm) thick and shall measure 0.3125 inch (7.938 mm) across the hexagonal flats. Thread size shall be 1/4-32NEF-2B. An internal-tooth lock washer shall be supplied and when mounted, shall have a maximum thickness of approximately 0.045 inch (1.14 mm). If retainer rings are used, they shall be not thicker than 0.032 inch (0.81 mm).

3.15 Pure tin. The use of pure tin, as an underplate or final finish is prohibited both internally and externally. Tin content of resistor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.4).

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.020	0.51	.187	4.75	.650	16.51
.002	0.05	.031	0.79	.250	6.35	.870	22.10
.005	0.13	.050	1.27	.310	7.87		
.010	0.25	.062	1.57	.380	9.65		
.015	0.38	.125	3.18	.500	12.70		

NOTE:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style RK09 resistor.

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TABLE I. Style RK09 resistor.

Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal	Dash number	
						Locking bushing	Standard bushing
RK09UACS100		10	.66	3.8		0001	
RK09UACS150		15	.88	4.9		0002	
RK09UACS200		20	.89	5.7		0003	
RK09UACS250		25	.80	6.2		0004	
RK09UACS350		35	.72	7.3		0005	
RK09UACS500		50	.65	8.6		0006	
RK09UACS750		75	.55	10		0007	
RK09UACS101		100	.68	12		0008	
RK09UACS151		150	.41	14		0009	
RK09UACS201		200	.55	17.4		0010	
RK09UACS251		250	.39	19		0011	
RK09UACS351		350	.36	23		0012	
RK09UACS501		500	.42	27		0013	
RK09UACS751		750	.35	32		0014	
RK09UACS102	.500	1,000	.34	39	S	0015	NA
RK09UACS152		1,500	.31	46		0016	
RK09UACS202		2,000	.23	55		0017	
RK09UACS252		2,500	.21	61		0018	
RK09UACS352		3,500	.21	73		0019	
RK09UACS502		5,000	.17	86		0020	
RK09UACS752		7,500	.21	104		0021	
RK09UACS103		10,000	.15	122		0022	
RK09UACS153		15,000	.13	150		0023	
RK09UACS203		20,000 3/	.12	173		0024	
RK09UACS253		25,000	.114	194		0025	
RK09UACS353		35,000	.102	229		0026	
RK09UACS503		50,000	.080	272		0027	
RK09UACS753		75,000 4/	.060	290		0190	
RK09UACS104		100,000 4/	.045	300		0191	
RK09SACS100		10	.66	3.8			0028
RK09SACS150		15	.88	4.9			0029
RK09SACS200		20	.89	5.7			0030
RK09SACS250		25	.80	6.2			0031
RK09SACS350		35	.72	7.3			0032
RK09SACS500		50	.65	8.6			0033
RK09SACS750		75	.55	10			0034
RK09SACS101		100	.68	12			0035
RK09SACS151		150	.41	14			0036
RK09SACS201	.500	200	.55	17.4	S	NA	0037
RK09SACS251		250	.39	19			0038
RK09SACS351		350	.36	23			0039
RK09SACS501		500	.42	27			0040
RK09SACS751		750	.35	32			0041
RK09SACS102		1,000	.34	39			0042
RK09SACS152		1,500	.31	46			0043
RK09SACS202		2,000 3/	.23	55			0044
RK09SACS252		2,500	.21	61			0045
RK09SACS352		3,500	.21	73			0046

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal	Dash number	
						Locking bushing	Standard bushing
RK09SACS502	0.500	5,000	17	86	S		0047
RK09SACS752		7,500	.21	104			0048
RK09SACS103		10,000	.15	122			0049
RK09SACS153		15,000	.13	150			0050
RK09SACS203		20,000	.12	173			0051
RK09SACS253		25,000	.114	194			0052
RK09SACS353		35,000	.102	229			0053
RK09SACS503		50,000	.080	272			0054
RK09SACS753		75,000 4/	.060	290			0192
RK09SACS104		100,000 4/	.045	300			0193
RK09SBCS100	.625	10	.66	3.8	S	NA	0055
RK09SBCS150		15	.88	4.9			0056
RK09SBCS200		20	.89	5.7			0057
RK09SBCS250		25	.80	6.2			0058
RK09SBCS350		35	.72	7.3			0059
RK09SBCS500		50	.65	8.6			0060
RK09SBCS750		75	.55	10			0061
RK09SBCS101		100	.68	12			0062
RK09SBCS151		150	.41	14			0063
RK09SBCS201		200	.55	17.4			0064
RK09SBCS251		250	.39	19			0065
RK09SBCS351		350	.36	23			0066
RK09SBCS501		500	.42	27			0067
RK09SBCS751		750	.35	32			0068
RK09SBCS102		1,000	.34	39			0069
RK09SBCS152		1,500	.31	46			0070
RK09SBCS202		2,000	.23	55			0071
RK09SBCS252		2,500	.21	61			0072
RK09SBCS352		3,500	.21	73			0073
RK09SBCS502		5,000	.17	86			0074
RK09SBCS752		7,500	.21	104			0075
RK09SBCS103		10,000	.15	122			0076
RK09SBCS153		15,000	.13	150			0077
RK09SBCS203		20,000 3/	.12	173			0078
RK09SBCS253		25,000	.114	194			0079
RK09SBCS353		35,000	.102	229			0080
RK09SBCS503		50,000	.080	272			0081
RK09SBCS753		75,000 4/	.060	290			0194
RK09SBCS104		100,000 4/	.045	300			0195
RK09UDCS100	0.875	10	.66	3.8	S	0082	NA
RK09UDCS150		15	.88	4.9		0083	
RK09UDCS200		20	.89	5.7		0084	
RK09UDCS250		25	.80	6.2		0085	
RK09UDCS350		35	.72	7.3		0086	
RK09UDCS500		50	.65	8.6		0087	
RK09UDCS750		75	.55	10		0088	
RK09UDCS101		100	.68	12		0089	
RK09UDCS151		150	.41	14		0090	
RK09UDCS201		200	.55	17.4		0091	

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal	Dash number	
						Locking bushing	Standard bushing
RK09UDCS251		250	.39	19		0092	
RK09UDCS351		350	.36	23		0093	
RK09UDCS501		500	.42	27		0094	
RK09UDCS751		750	.35	32		0095	
RK09UDCS102		1,000	.34	39		0096	
RK09UDCS152		1,500	.31	46		0097	
RK09UDCS202		2,000	.23	55		0098	
RK09UDCS252		2,500	.21	61		0099	
RK09UDCS352		3,500	.21	73		0100	
RK09UDCS502	.875	5,000	.17	86	S	0101	NA
RK09UDCS752		7,500	.21	104		0102	
RK09UDCS103		10,000	.15	122		0103	
RK09UDCS153		15,000	.13	150		0104	
RK09UDCS203		20,000 3/	.12	173		0105	
RK09UDCS253		25,000	.114	194		0106	
RK09UDCS353		35,000	.102	229		0107	
RK09UDCS503		50,000	.080	272		0108	
RK09UDCS753		75,000 4/	.060	290		0196	
RK09UDCS104		100,000 4/	.045	300		0197	
RK09SDCS100		10	.66	3.8			0109
RK09SDCS150		15	.88	4.9			0110
RK09SDCS200		20	.89	5.7			0111
RK09SDCS250		25	.80	6.2			0112
RK09SDCS350		35	.72	7.3			0113
RK09SDCS500		50	.65	8.6			0114
RK09SDCS750		75	.55	10			0115
RK09SDCS101		100	.68	12			0116
RK09SDCS151		150	.41	14			0117
RK09SDCS201		200	.55	17.4			0118
RK09SDCS251		250	.39	19			0119
RK09SDCS351		350	.36	23			0120
RK09SDCS501		500	.42	27			0121
RK09SDCS751		750	.35	32			0122
RK09SDCS102	.875	1,000	.34	39	S	NA	0123
RK09SDCS152		1,500	.31	46			0124
RK09SDCS202		2,000	.23	55			0125
RK09SDCS252		2,500	.21	61			0126
RK09SDCS352		3,500	.21	73			0127
RK09SDCS502		5,000	.17	86			0128
RK09SDCS752		7,500	.21	104			0129
RK09SDCS103		10,000	.15	122			0130
RK09SDCS153		15,000	.13	150			0131
RK09SDCS203		20,000 3/	.12	173			0132
RK09SDCS253		25,000	.114	194			0133
RK09SDCS353		35,000	.102	229			0134
RK09SDCS503		50,000	.080	272			0135
RK09SDCS753		75,000 4/	.060	290			0198
RK09SDCS104		100,000 4/	.045	300			0199

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal	Dash number	
						Locking bushing	Standard bushing
RK09SGCS100		10	.66	3.8			0136
RK09SGCS150		15	.88	4.9			0137
RK09SGCS200		20	.89	5.7			0138
RK09SGCS250		25	.80	6.2			0139
RK09SGCS350		35	.72	7.3			0140
RK09SGCS500		50	.65	8.6			0141
RK09SGCS750		75	.55	10			0142
RK09SGCS101		100	.68	12			0143
RK09SGCS151		150	.41	14			0144
RK09SGCS201		200	.55	17.4			0145
RK09SGCS251		250	.39	19			0146
RK09SGCS351		350	.36	23			0147
RK09SGCS501		500	.42	27			0148
RK09SGCS751		750	.35	32			0149
RK09SGCS102	0.125	1,000	.34	39	S	NA	0150
RK09SGCS152		1,500	.31	46			0151
RK09SGCS202		2,000	.23	55			0152
RK09SGCS252		2,500	.21	61			0153
RK09SGCS352		3,500	.21	73			0154
RK09SGCS502		5,000	.17	86			0155
RK09SGCS752		7,500	.21	104			0156
RK09SGCS103		10,000	.15	122			0157
RK09SGCS153		15,000	.13	150			0158
RK09SGCS203		20,000 3/	.12	173			0159
RK09SGCS253		25,000	.114	194			0160
RK09SGCS353		35,000	.102	229			0161
RK09SGCS503		50,000	.080	272			0162
RK09SGCS753		75,000 4/	.060	290			0200
RK09SGCS104		100,000 4/	.045	300			0201
RK09SKCS100		10	.66	3.8			0163
RK09SKCS150		15	.88	4.9			0164
RK09SKCS200		20	.89	5.7			0165
RK09SKCS250		25	.80	6.2			0166
RK09SKCS350		35	.72	7.3			0167
RK09SKCS500		50	.65	8.6			0168
RK09SKCS750		75	.55	10			0169
RK09SKCS101		100	.68	12			0170
RK09SKCS151		150	.41	14			0171
RK09SKCS201	2.500	200	.55	17.4	S	NA	0172
RK09SKCS251		250	.39	19			0173
RK09SKCS351		350	.36	23			0174
RK09SKCS501		500	.42	27			0175
RK09SKCS751		750	.35	32			0176
RK09SKCS102		1,000	.34	39			0177
RK09SKCS152		1,500	.31	46			0178
RK09SKCS202		2,000	.23	55			0179
RK09SKCS252		2,500	.21	61			0180
RK09SKCS352		3,500	.21	73			0181

See footnotes at end of table.

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TABLE I. Style RK09 resistor - Continued.

Type designation 1/	Shaft length (slotted shaft style) in inches ±.031	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal	Dash number	
						Locking bushing	Standard bushing
RK09SKCS502	2.500	5,000	.17	86	S	NA	0182
RK09SKCS752		7,500	.21	104			0183
RK09SKCS103		10,000	.15	122			0184
RK09SKCS153		15,000	.13	150			0185
RK09SKCS203		20,000 3/	.12	173			0186
RK09SKCS253		25,000	.114	194			0187
RK09SKCS353		35,000	.102	229			0188
RK09SKCS503		50,000	.080	272			0189
RK09SKCS753		75,000 4/	.060	290			0202
RK09SKCS104		100,000 4/	.045	300			0203

1/ For reference purposes only.
2/ Percent of resolution = 1/N x 100. N = number of turns.
3/ Maximum value based on use of wire having no less than .001 inch (0.13 mm) nominal diameter.

NOTE: For Navy use, resistance values shall be based on the use of wire having no maximum less than 0.001 inch (0.03 mm) wire diameter.

4/ Valued based on use of wire having no less than 0.0006 inch (0.15 mm) nominal diameter.

NOTE: For Army and Air Force use, resistance values shall be based on the of wire having no less than 0.0006 inch (0.15 mm) wire diameter.

3.16 Marking. Due to size limitations, marking of the schematic diagram is optional.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-39002.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-39002 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this date of this specification, the applicable associated specification, and the complete PIN.

b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.

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c. Packaging requirements (see 5.1).

6.3 Type designation. The type designation for identifying these parts is as follows:

NOTE: This is for information only. For the correct PIN, see 1.2.

RK09	SA	C	S	101
Style (6.3.1)	Shaft and type mounting (6.3.2)	Resistance temperature characteristic (6.3.3)	Terminal (6.3.4)	Resistance (6.3.5)

6.3.1 Style. The style is identified by the two letter symbol “RK” followed by a two digit number; the letters identify semi-precision wire-wound, variable resistors, and the number identified the physical size.

6.3.2 Shaft and type mounting. The style of shaft, type of mounting, and length of shaft is identified by a two letter symbol. The first letter indicates the style of shaft and type of mounting and the second letter indicates the length of shaft, see MIL-PRF-39002.

6.3.3 Resistance temperature characteristic. The resistance temperature characteristic of the completed resistor is identified by a single letter, see MIL-PRF-39002.

6.3.4 Terminals. The terminals are identified by a single letter, see MIL-PRF-39002.

6.3.5 Resistance. The nominal total resistance value expressed in ohms is identified by a three digit number; the first two digits represent significant figures and the last digit specified the number of zeros to follow, see MIL-PRF-39002.

6.4 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - CR
DLA - CC

Preparing activity:
DLA - CC

Review activity:
Army - AR

(Project 5905-2010-024)

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